

Autodesk® Scaleform®

AMP

Scaleform

(AMP)

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2012 6 15

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Autodesk® Scaleform® 4.2

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AMP

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1.2	AMP	2
2	AMP	7
2.1		7
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5.2		20
5.3		21
5.4		21
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7		23

1

AMP™

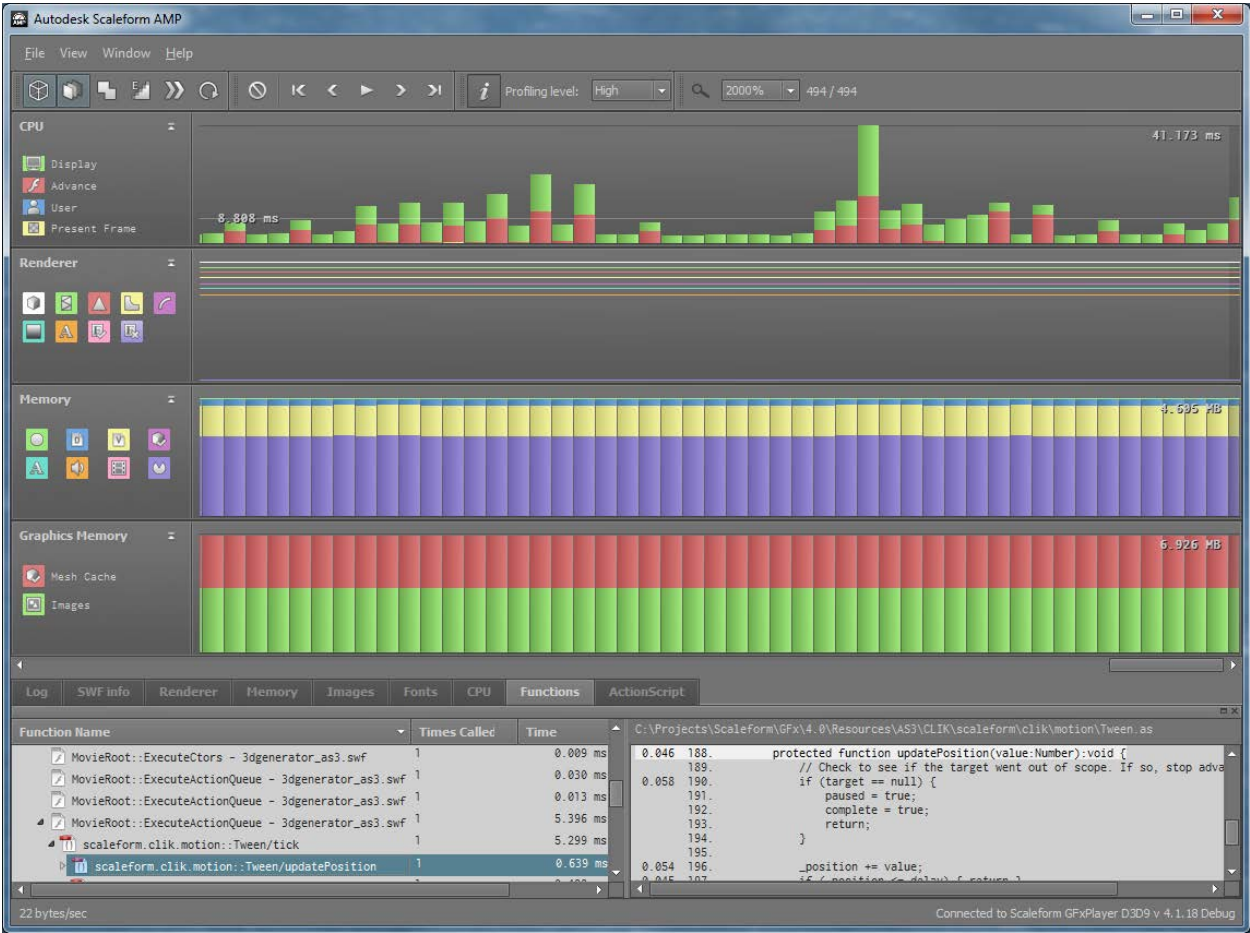
Scaleform®

CPU

Scaleform

AMP

ActionScript



1 Scaleform AMP

Scaleform

AMP

AMP

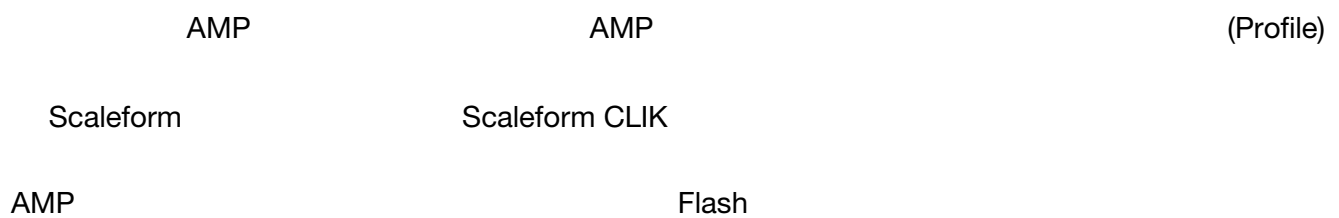
AMP

AMP Scaleform 3.2

1.1 AMP



1.2 AMP



- CPU (User) ActionScript (Advance) Scaleform (Display) Direct Access API Scaleform
CPU (PresentFrame)

- [Scaleform](#)

Scaleform

- Scaleform

- Scaleform AMP AMP
GPU

- “ ” “ ”(Log)

Scaleform

- “SWF ”(SWF info) flash
flash

- “ ”(Renderer)

- “ ”(Memory) : Scaleform Scaleform
AMP
“”

- “ ”(Images)

Scaleform

- AMP DXT SWF
“ ”(Fonts)

- CPU “ ”(Advance) “ ”(Display) “ ”(User)

- “ ”(Functions)

ActionScript C++

ActionScript VM

- **ActionScript** “ ”(Functions)
C++

- “ ”(Source) “ ” ActionScript AS3
AS SWF AS3
AS2
“ ”(high) AS2 Scaleform

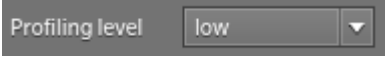
AMP SWF AS3 Flash SWF AMP

- **(File)**
 - o **(Connection)**
 - o **(Disconnect)**
 - o **(Debug Info Paths)**
AMP SWD AS2 AS AS3 “ ”(Add)
 - o “ ”(Remove)
 - o **(Load Profile Frames)**
 - o **(Save Profile Frames)**
 - o **(Dump Memory Report)** AMP
“AmpMemReport.txt”
 - o **(Exit)**
- “ ”(View)
 - o **(Graph Tooltips)** AMP
 - o **(Time Units)** “ ”
“ ”
 - o **(Memory Units)** “ ” “ ”

- “ ”(Window)
 - CPU
 - ActionScript
 - SWF
 - Restore UI
 - UI
- “ ”(Help)
 - AMP
 - “ ”(User Guide) --
 - Scaleform.com
 - “ ”(About) --
 - AMP
- - UI
 - 0  **(Wireframe)**
 - 0 
 - AMP
 - 0 
 - 0  **(Anti-aliasing)**
 - Scaleform
 - (None)
 - (EdgeAA)
 - (HW FSAA)
 - 0  **(Fast-forward)**
 - Flash
 - SWF
 - FPS
 - 0  **(Restart)**
 - Flash
 - “ ”(Frame control)
 - 0  **(Clear)**
 - AMP
 - 0  **(First Frame)**
 - ,
 - 0  **(Previous Frame)**
 - ,

- 0  (Pause)
- 0  (Next Frame)
- 0  (Last Frame)

- “ ”(Profiling)

- 0  (Detailed Profiling)
- “ ” AMP
- “ ” AMP
- “ ”
- 0  : “ ”(Low)
- “ ”(Medium) “ ”(High) AMP C++
- ActionScript C++
- ActionScript 2 (SWD)
- “ ”(Code)
- ActionScript 3
- 0  “ ”(Zoom)
- (Zooming out)
- (Zooming In)

- “ ”(Status)

AMP

UI AMP “ ”

“ UI”(Reset UI)

2 AMP

AMP

Scaleform

Bin/AMP

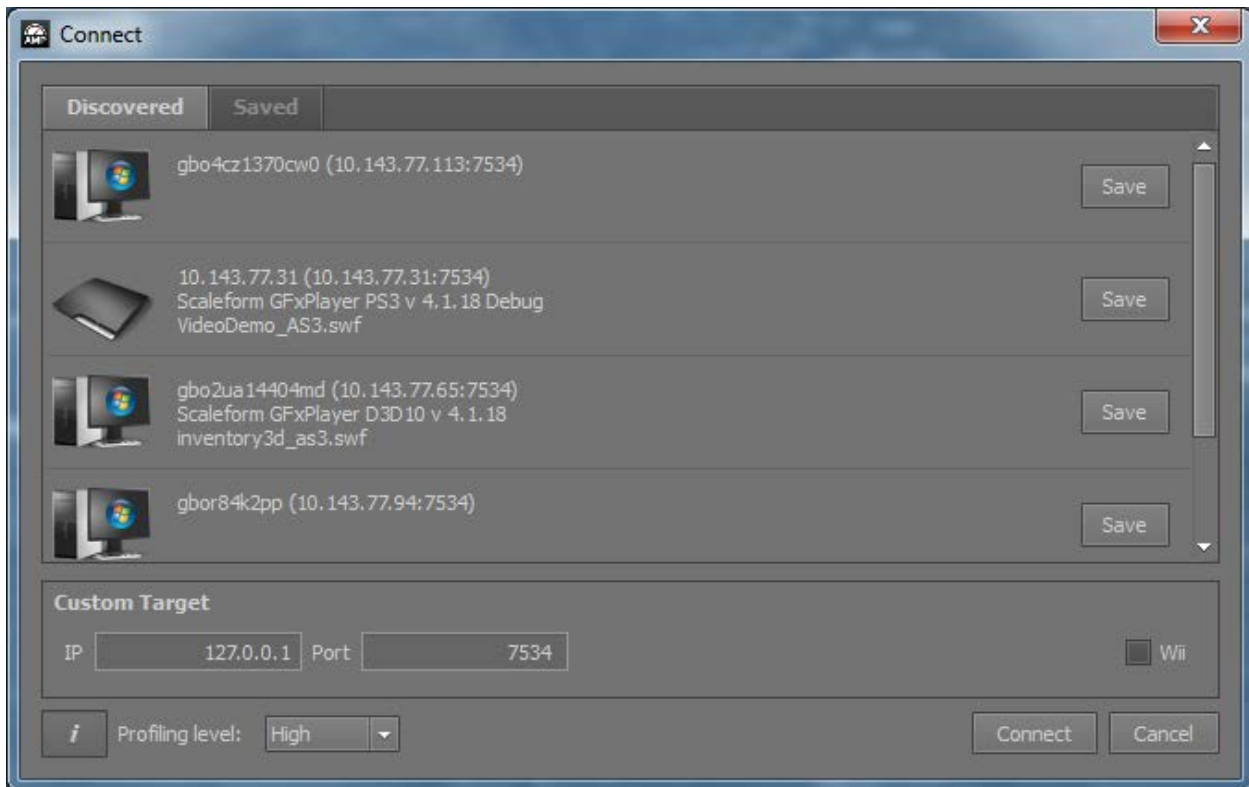
GFxAmpClient.exe

2.1

IP

AMP

AMP



1 AMP

“ ” (File) “ ” (Connection)

AMP

PC Xbox360 PS3
Flash

IP

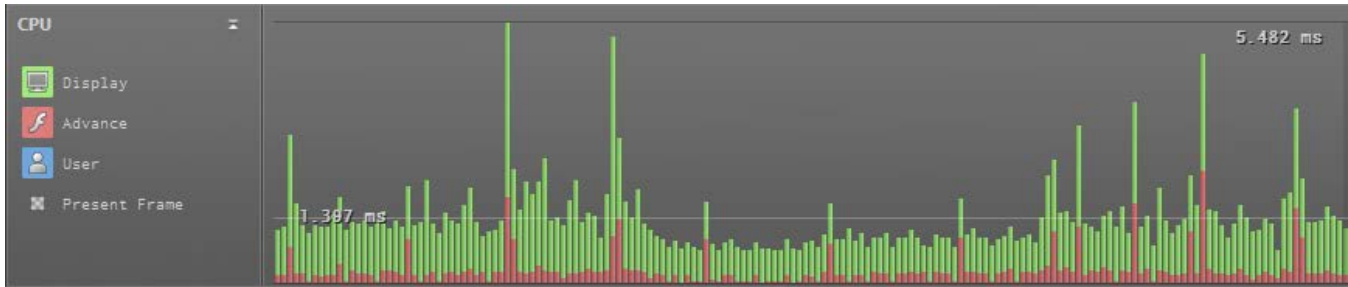
AMP

Scaleform AMP
“ ” (Connect)

Scaleform
Wii (NDEV) NDEV
USB COM Wii PC IP “wii”
Wii AMP

2.2

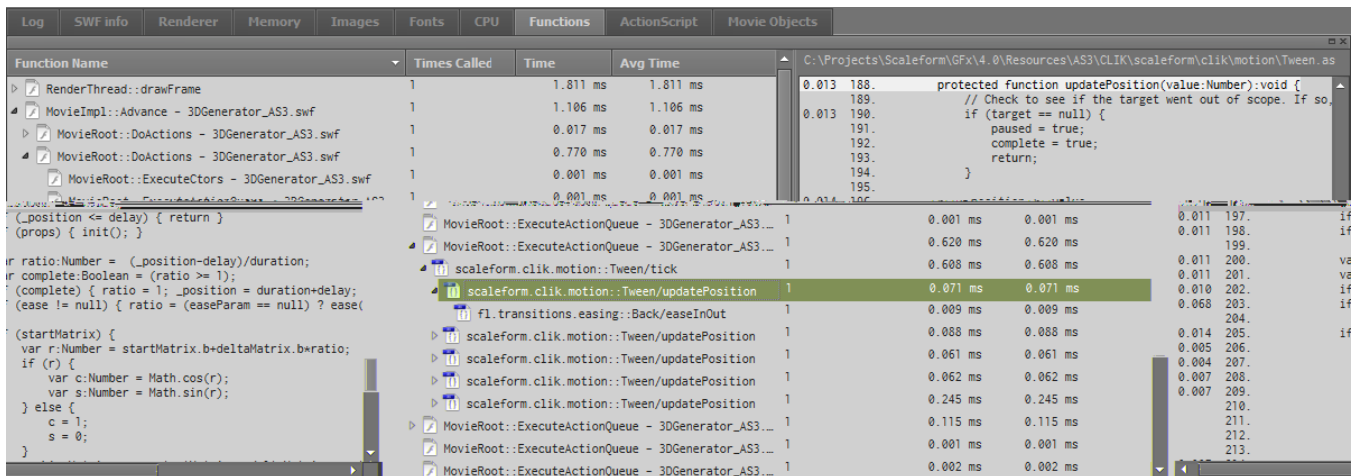
Scaleform (Display) ActionScript (Advance) Direct Access API (User) CPU



2 CPU

CPU
UI

CPU “CPU ” (CPU Summary) “ ” (Functions)
ActionScript “ CPU ”
ActionScript



3 CPU

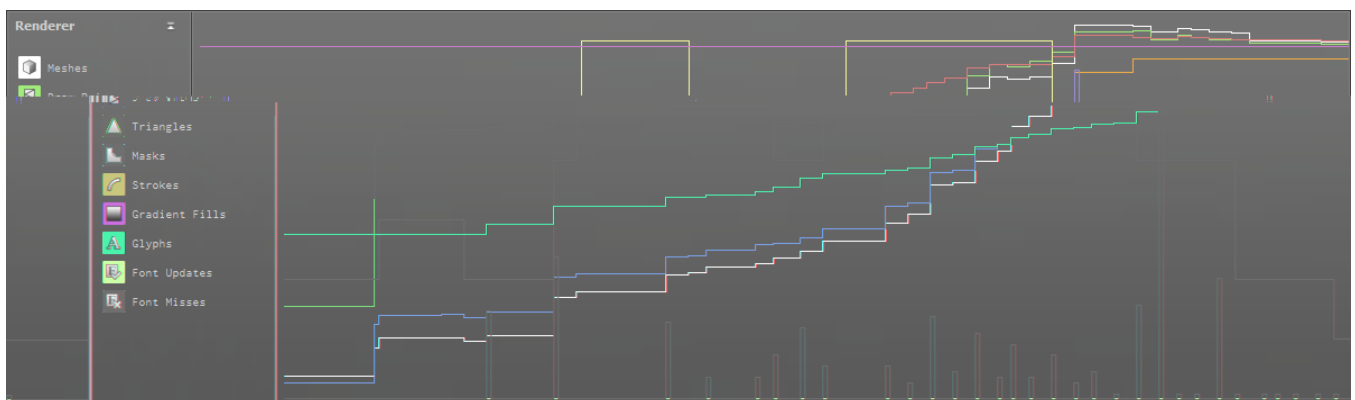
CPU

CPU

2.3

(Best Practices

Guide)



4

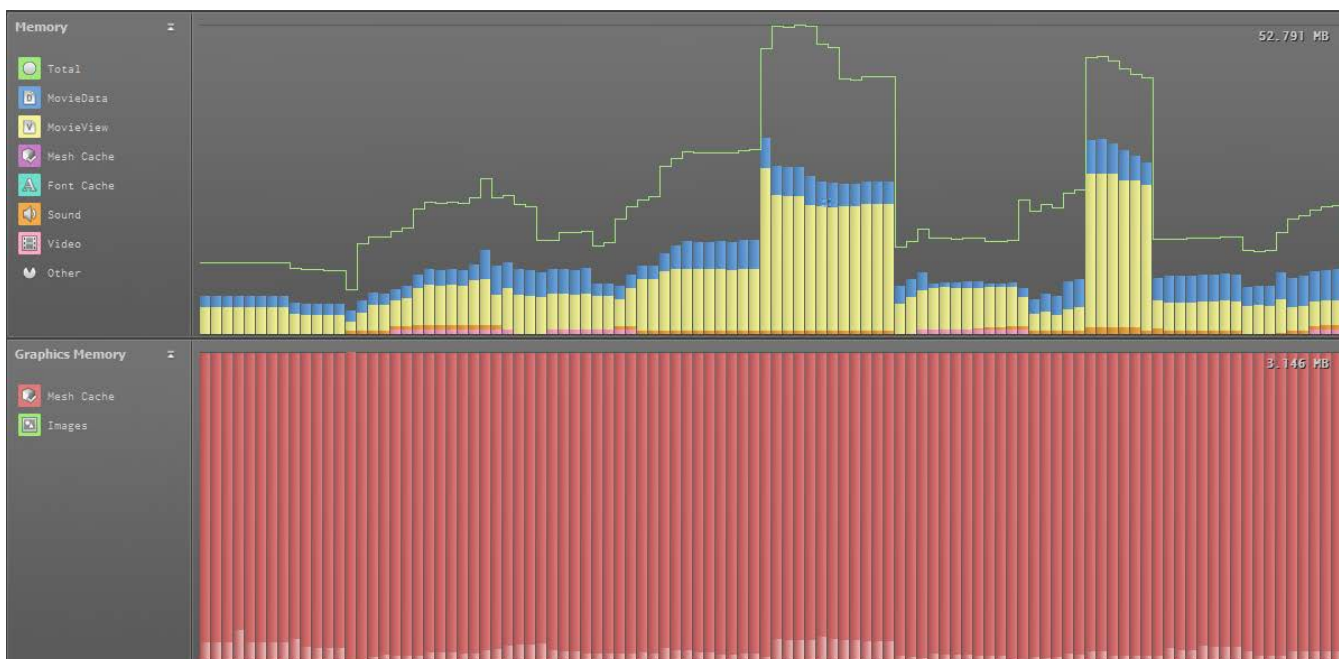
“ ” (Render Summary)

Log	SWF info	Renderer	Memory
Meshes:		182	
Draw Primitives:		183	
Triangles:		5124	
Masks:		6	
Filters:		0	
Strokes:		21	
Gradient Fills:		0	
Rasterized Glyphs:		412	
Font Cache Updates:		0	
Font Textures:		1	
Font Cache Thrashing:		0	
Font Cache Fill:		28%	
Font Cache:		1048576 pixels	
Font Cache Used:		298294 pixels	
Font Failures:		0	
Font Cache Misses:		5	
Mesh Thrashing:		0	

5

2.4

Scaleform

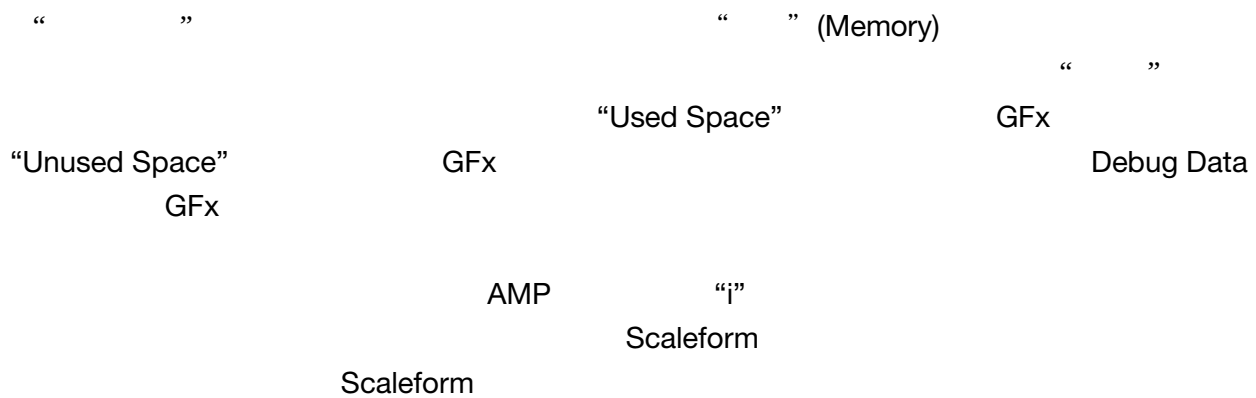


6

- **(Total)** Scaleform
- **(MovieData)** SWF GFX
- **(Movie view)** GfX::Movie
ActionScript
- **(MeshCache)**
(Tesselator)

- **(Font Cache)**
- **(Sound)**
- **(Video)**
- **(Other)** Scaleform
- **(Images)**

[\(Memory System Overview\)](#)



- o **Renderer** Scaleform
 - **Buffers** (Mesh Staging Buffer)
 - **RenderBatch**
 - **Primitive**
 - **Fill**

- **Mesh**
 - **MeshBatch** “ ”
 - **Context**
 - **NodeData**
 - **TreeCache**
 - **TextureManager**
 - **MatrixPool**
 - **MatrixPoolHandles**
 - **Text** [html](#)
 - **Font**
- o **Model** (dl)-5(e)De(Fof(nt)TETD 0 612 792 reW nBT/CS0 CS 0 SCN0.315 w /C2_1 1 Tf0 Tc 0 Tw 2 Tr 1

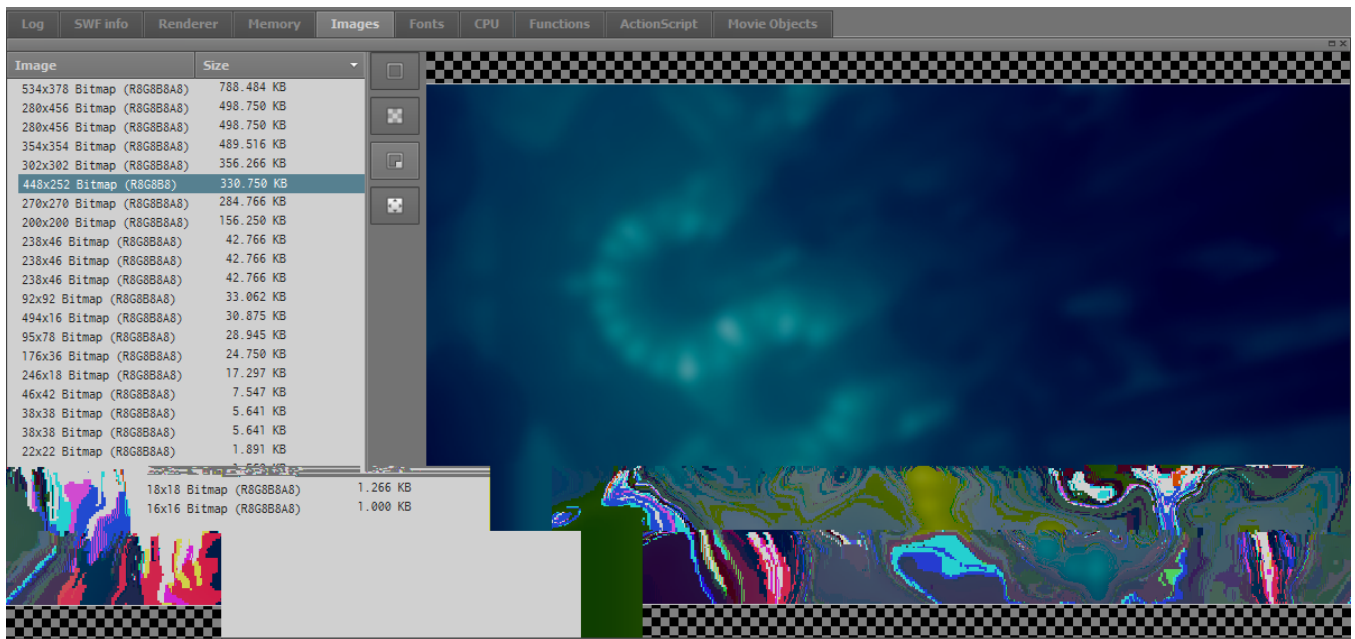
- **Vtable** **ActionScript**
- **SlotInfo** **ActionScript**
 - ClassTraits** **InstanceTraits** **SlotInfo** “ ” (Class)
 - “ ” (Instance) **ClassTraits** **InstanceTraits**
- o **SlotInfoHash** **ActionScript**
 - (Hash Table)
- **ClassTraits** **ActionScript**
- **Class** **ActionScript**
- **InstanceTraits** **ActionScript**
- **Instance** **ActionScript**
- **AbcFile** **ABC**
 - o **AbcConstPool** **ABC**
 - Multiname**
 - AbcFile**
- **VMAbcFile** **ABC**
 - AbcFile**
- **Tracer**
- o **IME** (IME)

Log	SWF info	Renderer	Memory	Image
Memory			Size	
▲	Total Footprint		4849.402 KB	
▲	Used Space		4633.715 KB	
▷	Video Heaps		0.000 KB	
▷	Other Heaps		325.621 KB	
▲	Movie View Heaps		1226.094 KB	
▲	MovieView "3dgenerator_as3.swf"		1226.094 KB	
	String		0.828 KB	
▷	Renderer		12.672 KB	
▷	MovieView		1176.781 KB	
	General		35.812 KB	
▷	Movie Data Heaps		261.066 KB	
▲	Global Heap		2820.934 KB	
	Video		264.176 KB	
	String		20.633 KB	
	Sound		0.141 KB	
▷	Renderer		2310.914 KB	
▷	MovieView		8.039 KB	
▷	MovieDef		116.676 KB	
	Image		0.141 KB	
	IME		9.914 KB	
	General		90.301 KB	
▷	Debug Memory		8.000 KB	
▷	Unused Space		38.820 KB	
	Debug Data		176.867 KB	

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Scaleform

“ ” (Images)



9 “ ”

Scaleform

Scaleform

“ ” (Fonts)



10 “ ”

SWF

AMP

MovieData

CharDefs

“ ”

“ ”

SWF

2.5

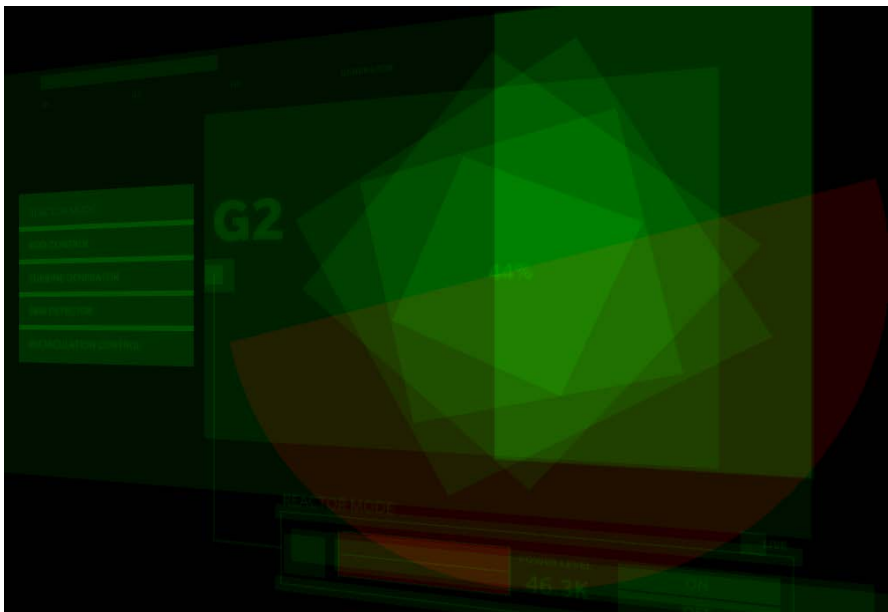
AMP

(Highlight)

11

16

8



12

2.6

AMP

3

1. AMP GfX::System GfX::System::Init
 Saleform::System Scaleform
 GfX::AMP::Server::Init() GfX::AMP::Server::Uninit() AMP
2. AMP Windows Xbox360 PS3 Linux Mac, Wii, Android iPhone/iPad,
 3DS, PS Vita, WiiU AMP
 AMP IP IP
- Xbox360 IP IP IP Xbox
 360
- PS3 IP IP
- IP
- Android
 AndroidManifest.xml
 <uses-permission android:name="android.permission.INTERNET" />
- PC WiFi WiiU PC 6003
 TCP/IP
- Wii AMP NDEV USB COM
 PC IP "wii"
 Wii AMP RVL_SDK/X86/bin
 AMP hio2.dll
3. Unreal Engine 3 AMP
 Flash AS2 SWD SWD AMP
 AMP SWF

	AMP	Scaleform	Scaleform	AMP
1.	GFxConfig.h	SF_AMP_SERVER	Scaleform	Scaleform
	<pre>Ptr<GFx::AMP::ServerState> serverState = *SF_NEW GFx::AMP::ServerState(); serverState->StateFlags &= GFx::AMP::Amp_Disabled; AmpServer::GetInstance().UpdateState(serverState);</pre>			
				AMP
2.	AMP	(Network Socket)		
		Ws2_32.lib	Windows Xnet.lib	Xbox360 libnetctl
	PS3	Linux	.	
3.	AMP			
		---		AMP
				AMP
		AMP::Server::GetInstance().SetInitSocketLib(false)		
		AMP		
4.	AMP		Scaleform	
		AMP::Server::GetInstance().SetHeapLimit()		
5.	AMP	7534		
	AMP::Server::GetInstance().SetListeningPort()			
6.	AMP			
	AMP::Server::GetInstance().SetConnectionWaitTime			

5 AMP

Scaleform
Scaleform Player AMP AMP

5.1 Debug

AMP ActionScript 2 Flash SWD
Flash CS3 CS4 (Ctrl+Shift+Enter) SWD SWD
SWF AMP

5.2

AMP AMP

AMP

- AMP::AppControlInterface
HandleAmRequest AMP AMP::MessageAppControl
- AMP::Server::GetInstance().SetAppControlCallback
- AMP::Server::GetInstance().SetAppControlCaps AMP
AMP::MessageAppControl True

```
AMP::MessageAppControl caps;  
caps.SetCurveToleranceDown(true);  
caps.SetCurveToleranceUp(true);  
caps.SetNextFont(true);  
caps.SetRestartMovie(true);  
caps.SetToggleAaMode(true);  
caps.SetToggleAmpRecording(true);  
caps.SetToggleFastForward(true);  
caps.SetToggleInstructionProfile(true);  
caps.SetToggleOverdraw(true);
```



```
caps.SetToggleBatch(true);
caps.SetToggleStrokeType(true);
caps.SetTogglePause(true);
caps.SetToggleWireframe(true);
AMP::Server::GetInstance().SetAppControlCaps(&caps);
```

5.3

AMP

AMP::Server::GetInstance().SetConnectedApp

AMP

AMP

- AMP::Server::GetInstance().SetState ServerStateType
 - o Amp_RenderOverdraw
 - o Amp_App_Wireframe
 - o Amp_App_Paused
 - o Amp_App_FastForward
- AMP::Server::GetInstance().SetAaMode
- AMP::Server::GetInstance().SetStrokeType
- AMP::Server::GetInstance().SetCurrentLocale
- AMP::Server::GetInstance().SetCurveTolerance
- AMP::ServerState
 - AMP::Server::GetInstance().UpdateState

5.4

AMP

CPU

C++

ActionScript

ActionScript

Amp.addMarker(1)

Amp.addMarker() .as

Resources/AS2/CLIK

Amp

C++

GFx::MovieImpl:: AdvanceStats->AddMarker(1)

AMP

6 AMP

AMP

1. 游戏端使 AMP 工作所需的最低限度是什么？

AMP GFX_AMP_SERVER GFx
AMP Platform::RenderThread::drawFrame
AMP AmpServer::GetInstance().AdvanceFrame

2. 使用 AmpClient 连接我们的游戏时，我们的游戏的帧速率变得非常缓慢。我们发现 AmpServer::AdvanceFrame() 花费 时间。

AMP
AMP
AMP
AMP
“i” |

3. Windows/PS3/Xbox AMP

AMP
Platform::RenderThread::drawFrame AMP
AmpServer::GetInstance().AdvanceFrame

4. AMP 为何在更新时暂停一秒钟或更长时间 (AMP::Server::WaitForAmpConnection) ?

AMP AMP Scaleform
AMP AMP 'i'
AMP ' (profile level)
1MB
GFx::AMP::Server::SetHeapLimit

- 5.

AMP 1MB
AmpServer::SetHeapLimit
GfX

AMP

7

AMP

- [AMP](#) (AMP Use Cases)
- [Scaleform Reference](#) – /
- [AMP Forum](#) –